

DRONESMS – LOVABLE BUILD PROMPT

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Build a SaaS web application called DroneSMS. This is a Safety Management System platform for commercial drone operators in the United States. Operators subscribe, document jobs, manage crew training, and export professional PDF compliance packets.

The app must work excellently on an iPad in Safari. This is the primary use case. Mobile responsive throughout.

TECH STACK

- React frontend
- Supabase for auth and database
- Stripe for subscription billing
- PDF generation for compliance packet export
- Tailwind CSS for styling
- Clean, light, professional UI – white backgrounds, navy sidebar, blue accents

AUTHENTICATION

- Email and password registration
- Email verification required before dashboard access
- Login with redirect to dashboard
- If user has no company profile after login, redirect to onboarding screen first
- Role-based access: operator role and admin role

DATABASE TABLES (Supabase)

users

- id, email, full_name, role (operator/admin), created_at, email_verified

companies

- id, user_id, company_name, business_type, state, part107_cert, drone_platforms, setup_complete, created_at

jobs

- id, company_id, client_name, site_address, job_type, scheduled_date, drone_platform, crew_names (json), status (Planning/Active/Closed), created_at

template_versions
- id, template_slug, version_number, field_schema (json), published_at, is_latest

template_completions
- id, job_id, template_slug, template_version, field_data (json), status
(incomplete/in_progress/complete), completed_at, completed_by

training_records
- id, company_id, employee_name, role, training_date, training_type,
training_method, trainer_name, notes, acknowledged, acknowledged_at, created_by

subscriptions
- id, user_id, stripe_customer_id, stripe_subscription_id, tier
(field_operator/company_standard), status, current_period_end

export_logs
- id, job_id, exported_at, exported_by, template_versions (json), storage_path

announcements
- id, content, published_at, published_by, active

SCREENS TO BUILD

1. REGISTRATION SCREEN /register

Fields: full name, email, password, confirm password

Submit creates user, sends verification email, redirects to verification pending screen

Duplicate email shows inline error

2. LOGIN SCREEN /login

Fields: email, password

Forgot password link

On success: check email verified, check company profile complete, redirect accordingly

3. COMPANY PROFILE SETUP /onboarding/company

Required before dashboard. Gate – dashboard checks this on every load.

Fields: company name, business type (dropdown), state, FAA Part 107 certificate number, drone platforms (text)

Submit writes company record, redirects to dashboard

4. DASHBOARD /dashboard

Stats row: active jobs count, packets exported this month, training records count

Active jobs list preview (max 5) with client name, date, completion badge

Quick action buttons: New Job, Add Training Record
Admin announcement strip if active announcement exists
Warning banner if COI expiring (future feature placeholder – show static for now)

5. JOB LIST /jobs

Search bar by client or address
Filter by status and job type
Table: client name, address, job type badge, date, progress bar (X/6), status badge, Open button
New Job button top right

6. CREATE JOB /jobs/new

Form sections: Site and Client Info, Crew Entry, Notes
Fields: client name, job type (dropdown), site address, scheduled date, drone platform, crew rows (name + role dropdown), notes (optional)
Crew rows: add and remove dynamically
On submit: create job record + create 6 template_completion records with status incomplete linked to job
Redirect to job file hub on success

7. JOB FILE HUB /jobs/:id

Job header: client name, address, date, status badge
Meta row: client, drone, crew, job ID
Progress bar: X of 6 templates complete
Template checklist – 6 rows in this order:
 01 Job Hazard Analysis (jha)
 02 Pre-Flight Checklist (preflight)
 03 Crew Briefing Sign-Off (crew-briefing)
 04 LAANC Authorization Log (laanc)
 05 Incident / No-Incident Closeout (incident)
 06 Training Records – Crew (training-summary)
Each row shows: number, icon, template name, last edited or completed info, status badge, Fill Out or View button
Export Compliance Packet button at bottom – disabled with tooltip if any template incomplete, active when all 6 complete
Completed templates are fully read-only

8. TEMPLATE SCREENS /jobs/:id/templates/:slug

Build all 6 templates as fillable web forms. Each saves field_data to template_completions table on submit.
Auto-save draft on field blur. Final submit locks template as complete and read-only.
Template version recorded at time of first save.

TEMPLATE 1: JHA (jha)

Sections and fields:

- Site conditions: surface type (dropdown: concrete/glass/metal/other), building

height (number, feet), site access (text)

- Environmental conditions: wind speed (number, mph), weather (dropdown: clear/partly cloudy/overcast/light rain), visibility (dropdown: excellent/good/fair/poor)
- Hazard identification: repeating rows – hazard description (text), likelihood (1-5 dropdown), severity (1-5 dropdown), risk score (auto-calculated: likelihood x severity, display only), mitigation action (text). Add row button.
- If any risk score is 15 or higher show a warning banner: High risk hazard identified. Verify mitigations before proceeding.
- Nearby population: public presence (yes/no), exclusion zone planned (yes/no), exclusion zone description (text, shown if yes)
- Water and chemical: runoff risk (yes/no), chemical type (text, shown if yes), containment plan (text, shown if yes)
- Airspace: airspace class (dropdown: B/C/D/E/G), LAANC required (yes/no/not applicable)
- Overall risk rating: auto-calculated from highest hazard score – Low (1-8), Medium (9-14), High (15-25). Display prominently.
- Assessor name (text), assessment date (date, default today)
- Submit button

TEMPLATE 2: Pre-Flight Checklist (preflight)

Checkbox groups with Yes/No/NA options and notes field per item:

- Drone condition: visual inspection, props, motors, body
- Battery: charge level (number, %), cycle count (number), damage check
- Remote control: power, signal, controls
- Display/app: connection, telemetry, GPS lock
- Environmental: wind speed confirmed (number), no precipitation (yes/no), visibility confirmed (yes/no)
- Final go/no-go decision: radio buttons Go / No-Go. If No-Go selected, require notes field.
- Inspector name (text), timestamp auto-captured server-side on submit

TEMPLATE 3: Crew Briefing Sign-Off (crew-briefing)

- Briefing content checklist: checkboxes for – hazards reviewed, emergency procedures, exclusion zones, roles and responsibilities, stop-work authority, comms plan
- Crew acknowledgment section: pre-populated from job crew list. Each crew member row: name (read-only from job), role (read-only), acknowledgment checkbox, timestamp auto-captured when checked
- Operator can add additional crew not on original list
- All crew rows must be acknowledged before submit
- Briefing conducted by (text), briefing date and time
- Completed template is fully read-only

TEMPLATE 4: LAANC Authorization Log (laanc)

- Airspace class (dropdown: B/C/D/E/G)
- LAANC required (Yes/No/Not Applicable)

- If LAANC required = Yes: provider used (text), authorization number (required, text), authorization date (date), max altitude authorized (number), expiration date/time
- If airspace class B/C/D and LAANC = No: show warning banner (non-blocking)
- B4UFLY verified (Yes/No, required regardless of airspace class)
- Notes (text), operator name (text), date (date, default today)

TEMPLATE 5: Incident / No-Incident Closeout (incident)

- Incident occurred (Yes/No radio)
- If No: operator name (text), confirmation timestamp auto-captured, submit
- If Yes: incident type (dropdown), date and time of incident, location description (text), persons involved (text), description of what happened (textarea), immediate actions taken (textarea), equipment damage (yes/no + description if yes), injury (yes/no + description if yes), FAA reportable (yes/no, operator judgment), corrective actions planned (textarea), reporter name (text), date
- If incident_occurred = Yes, write incident_flag = true to job record

TEMPLATE 6: Training Records – Crew (training-summary)

- Read-only summary pulled from training_records table for crew members listed on this job
- Shows each crew member name, role, and their most recent training records
- Link to full Training Log
- Operator confirms crew training is current (checkbox required)
- Submit records confirmation

9. TRAINING LOG /training

Filter bar: search by employee name, training type dropdown, date range

Table: employee name, role, training type, date, method, trainer, acknowledged badge, View button

Export PDF button (exports filtered records)

Add Training Record button

10. CREATE TRAINING RECORD /training/new

Fields: employee name, role (dropdown), training date, training type (dropdown with these options: SMS Orientation / FAA Part 107 Requirements / Site-Specific Hazard Briefing / Drone Platform Operation / Equipment Inspection / Emergency Response / Water and Chemical Handling / PPE Requirements / Pressure Equipment Safety / Manufacturer Certification / Other), training method (dropdown: In Person / Video / Written Material / Hands-On / Combination), trainer name, notes (optional)

Employee acknowledgment checkbox: required. Label: I confirm I received and completed this training.

Submit writes training_record, acknowledged_at captured server-side

11. PDF EXPORT /jobs/:id/export

Triggered from Export button on Job File Hub when all 6 templates complete

Generate single PDF containing:

- Cover page: company name, job details, DroneSMS branding, template version, generation timestamp
- Required legal disclaimer on cover page (exact text): This compliance packet was generated using DroneSMS and is intended to support documentation of safety management practices. DroneSMS is not a regulatory authority. This document does not guarantee legal or regulatory compliance with FAA, OSHA, EPA, or any other governing body. Operators are responsible for verifying that their operations meet all applicable federal, state, and local requirements.
- Each of the 6 completed templates rendered in order
- Final page: job ID, export timestamp, operator name

Store PDF in Supabase storage. Write export_log record. Allow re-download from job file hub.

12. SUBSCRIPTION AND BILLING /account/billing

Two tiers:

Field Operator: \$49/month or \$499/year. 1 user. 6 core templates. Unlimited jobs. Training log. PDF export.

Company Standard: \$149/month or \$1499/year. Up to 5 users. Everything in Field Operator. Saved crew profiles.

Stripe Checkout for payment. Stripe webhooks for: subscription created, updated, deleted, payment failed, payment succeeded.

Current plan display with next billing date. Update payment method. Cancel option. Invoice history.

13. ACCOUNT SETTINGS /account/settings

Edit full name, email (re-verification if changed), password change (requires current password)

14. ADMIN: USER MANAGEMENT /admin/users

Admin role only. Search and filter users. Table: name, email, company, tier, status, signup date. Activate/deactivate toggle.

15. ADMIN: TEMPLATE EDITOR /admin/templates

Select template from dropdown. View current field schema. Edit field labels, types, required flags, dropdown options. Add or remove fields. Preview. Publish New Version button.

Publishing creates new template_version record with incremented version number. Existing template_completions remain locked to the version at time of completion. In-progress completions show banner: a new template version is available.

16. ADMIN: ANNOUNCEMENTS /admin/announcements

Text area for announcement. Publish button. Only one active announcement at a time. Clear button.

17. 404 AND ERROR SCREEN

Unauthenticated users redirect to login. Authenticated users see error with

dashboard link.

SHARED REQUIREMENTS

Navigation: fixed sidebar on desktop, hamburger on mobile. Nav items: Dashboard, Jobs, Training Log, Billing, Settings, Admin (admin role only).

Feature gating: Company Standard features (saved crew profiles, multiple users) blocked for Field Operator accounts with upgrade prompt.

All forms: validation before submit, inline error messages, loading states on submit buttons.

Mobile: all screens must work on 375px viewport minimum. Forms, checklists, and job creation must be fully usable on a tablet touchscreen.

Template version locking: when a template is submitted as complete, record the `template_version` at that moment. Completed templates always display the version they were submitted under regardless of future template updates.

Phase 2 data (capture now, do not display yet): template completion status per job, required training categories per crew member with completion flag, COI expiration date field on company profile, equipment inspection dates, incident closure status. These fields must exist in the schema even though the compliance health score dashboard is not built yet.

DESIGN SYSTEM

Font: Use a clean sans-serif. Syne for headings, DM Sans for body.

Colors:

Sidebar background: #162032

Primary blue: #1B5FE8

Background: #F7F8FA

White cards: #FFFFFF

Border: #E4E8EF

Text primary: #162032

Text secondary: #3D4F66

Text muted: #7A8FA8

Green: #0E9E5E

Amber: #C47A00

Red: #D93939

Status badges: colored pill badges — blue for active/planning, green for complete,

amber for in progress, gray for incomplete, red for incident flagged.

Cards: white background, 1px border, 10px border radius, subtle shadow.

Buttons: primary blue filled, secondary white with border, ghost transparent.

BUILD ORDER

1. Auth screens and Supabase connection
2. Company onboarding and dashboard
3. Job list and create job
4. Job file hub with template checklist
5. All 6 template forms
6. Training log and create training record
7. PDF export
8. Stripe billing
9. Admin screens
10. Mobile responsiveness pass

TEMPLATE 7: AGRICULTURAL COMPLIANCE ADDENDUM (ag-compliance)

Shown in job file hub only when job_type = Agriculture. Optional but strongly recommended.

Sections and fields:

- Application details: crop type (text), target pest or application purpose (text), total area treated (number, acres)
- Chemical product: product name (text), EPA registration number (text), application rate (number, per acre), formulation type (dropdown: liquid/granular/dust/other)
- Applicator credentials: applicator license number (text), license expiration date (date), issuing state (dropdown), license type (dropdown: commercial/private/certified/not required)
- Environmental conditions at time of application: wind speed (number, mph), wind direction (text), temperature (number, fahrenheit), relative humidity (number, percent)
- Buffer zones: buffer zones observed (yes/no), buffer distance from water bodies (number, feet), buffer distance from structures (number, feet), buffer distance from property lines (number, feet)
- Spray drift mitigation: nozzle type used (text), boom height (number, feet), droplet size classification (dropdown: very fine/fine/medium/coarse/very coarse), drift reduction technology used (yes/no + description)
- On-site documentation: SDS on site (yes/no), applicator license on site (yes/no), PPE worn by crew (multi-select checklist: gloves/eye

protection/respirator/coveralls/boots/none)

- Regulatory compliance: FIFRA compliance confirmed (yes/no checkbox), state ag department requirements met (yes/no checkbox), Clean Water Act buffer zones observed (yes/no checkbox)
- Reportable events: any reportable discharge or drift incident (yes/no), if yes: description (textarea), immediate actions taken (textarea), agency notified (text)
- Supervisor signature (text), application date (date, default today), application start time, application end time

TEMPLATE 8: CLEANING COMPLIANCE ADDENDUM (cleaning-compliance)

Shown in job file hub only when job_type = Commercial Cleaning or Residential Cleaning. Optional but strongly recommended.

Sections and fields:

- Cleaning agent: product name (text), manufacturer (text), SDS on site (yes/no), biodegradable (yes/no), EPA Safer Choice certified (yes/no), pH level (text)
- Water source: water source type (dropdown: municipal supply/tanked water/on-site reclaim/other), estimated water volume used (number, gallons)
- Wastewater management: estimated wastewater volume generated (number, gallons), discharge point (dropdown: contained and reclaimed/sanitary sewer with permit/third party disposal vendor/storm drain – NOTE: storm drain selection triggers warning banner: Storm drain discharge of wash water containing cleaning agents may violate Clean Water Act Section 402. Verify local permit requirements before proceeding.)
- Clean Water Act compliance: Clean Water Act compliance confirmed (yes/no checkbox with operator acknowledgment text: I confirm that wastewater discharge from this operation complies with applicable Clean Water Act requirements and local stormwater ordinances.), local municipal stormwater permit required (yes/no), permit number (text, shown if yes)
- Containment setup: runoff containment method (multi-select: ground tarps/vacuum recovery/containment berm/bladder tank/absorbent material/other), photos of containment setup taken (yes/no)
- Reclamation: reclamation method (dropdown: collection tank/vacuum recovery/absorbent material/third party vendor/not required), reclamation volume (number, gallons), disposal vendor name (text, shown if third party vendor selected), disposal vendor contact (text, shown if third party vendor selected)
- Surface details: surface type cleaned (multi-select: glass/concrete/metal/stone/painted surface/other), height of structure (number, feet), structure type (dropdown: commercial building/residential/industrial/infrastructure/other)
- OSHA compliance: SDS reviewed with crew (yes/no), PPE worn by crew (multi-select: gloves/eye protection/chemical-resistant footwear/coveralls/respirator/none), chemical handling training completed (yes/no)
- Supervisor signature (text), job date (date, default today)

JHA TEMPLATE ADDITIONS – Environmental and Chemical section (expand existing section):

Add these fields to the existing Water and Chemical section in Template 1:

- Regulatory citations applicable (multi-select: Clean Water Act Section 402/Clean Water Act Section 404/FIFRA/California DPR/Other State Ag Department/OSHA HazCom/Not Applicable)
- Water body proximity (yes/no), if yes: distance in feet (number), water body type (dropdown: river/stream/lake/pond/irrigation canal/storm drain/wetland/other)
- Secondary containment in place (yes/no)
- Reclamation method (dropdown: collection tank/absorbent material/containment berm/third party vendor/not required)
- Reclamation volume estimate (number, gallons)
- Disposal vendor name and contact (text, shown if third party vendor selected)

DATABASE ADDITIONS:

Add job_type values to jobs table: Agriculture, Commercial Cleaning, Residential Cleaning (in addition to existing: Inspection, Survey, Other)

Add to template_completions table: addendum_type field to track which optional addendum templates are associated with which job types.

BUILD ORDER ADDITIONS:

After step 5 (all 6 template forms), add:

5b. Agricultural compliance addendum (ag-compliance) – conditional on job type

5c. Cleaning compliance addendum (cleaning-compliance) – conditional on job type

TRAINING LOG ADDITIONS:

Add these training types to the dropdown in Create Training Record:

- Pesticide Applicator Certification
- Chemical Application Safety
- Stormwater and Clean Water Act Compliance
- FIFRA Compliance
- Wastewater Reclamation Procedures